
Dr. Hira Zaheer

Assistant Professor

Division of Mathematics

School of Basic Sciences

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TEACHING EXPERIENCE:

Galgotias University, Greater Noida, U.P

October 05, 2021- At Present

Assistant Professor

Netaji Subhas University of Technology, New Delhi

April 19, 2021- August 15, 2021

Guest Faculty

Jamia Millia Islamia University, New Delhi

August 19, 2019- June 30, 2020

Assistant professor (On contract)

Maharshi Markandeshwar University, Ambala, Haryana

July 01, 2017 – August 15, 2019

Assistant Professor

Shri Ram College Muzaffarnagar, U.P

January 6, 2012 - January 5, 2013

Assistant Professor

Ph.D: Topic of my research is *“Enhanced Differential Evolution Variants for Real Life Optimization Problems”*. Completed Courses for Ph.D. are as follows-

- 1. Evolutionary Algorithms**
- 2. Ordinary and Partial Differential Evolution**
- 3. Optimization Techniques**

The main work of my PhD is focused on the following two criteria.

- (i) Designing efficient algorithms
- (ii) Applying those algorithms for solving global real life optimization problems

In my work, I have proposed five variants of Differential Evolution in which four variants are for dealing with continuous optimization problems, while the fifth variant is for solving discrete optimization problems. The main steps in DE are initialization, mutation, crossover and selection. And the four continuous variants are proposed in such a way that there is in modification in each of the above listed steps for each variant respectively. The efficiency of proposed algorithms (continuous) was tested on three real life problems, one is of portfolio optimization and the other two are industrial optimization problems and one variant is tested on a real-life model of machine tool selection problem.

The main highlights of work are as follows:

- Studied Differential Evolution (DE) technique and developed a code in CPP and MATLAB
- Modified different variants of DE for discrete and continuous optimization problems
- Tested the efficiency of the proposed algorithms on various complex benchmark problems
- Compared the results with other popular meta-heuristics
- Applied those techniques on the real-life problems arising in industry and financial domain

EDUCATION:

Program	Year	Institution/Board	Percentage
Ph.D. (Mathematics)	2013-2017	Indian Institute of Technology Roorkee	--
M.Sc. (Mathematics)	2009-2011	D.A.V College, Muzaffarnagar, CCS University, Meerut	77.5
B.Sc (PCM)	2006-2009	SD Degree College Muzaffarnagar, CCS University, Meerut	73.5
Intermediate (PCM)	2005-2006	K.V O. F Muradnagar, CBSE	77.4
Highschool (Science)	2003-2004	K.V JLA Bareilly, CBSE	70.0

ACADEMIC ACHIEVEMENTS:

- Qualified National Eligibility Test (NET) JRF 2012 with **AIR- 95**
- Qualified Graduate Aptitude Test for Engineering (GATE) 2013 with **AIR-84**

COMPUTER PROFICIENCY:

- Operating System Windows, Linux, Mac
- Languages C, C++, MATLAB, SciLab, Python

PH.D SUPERVISION: Guiding one student Aiman Siddique in Galgotias University

Topic : A novel metaheuristic technique for real-life optimization problems.

University : Galgotias University Gr Noida

PROJECT GUIDED

Jan - July – 2023| **Student Name :** Chitrang Mishra

Topic: An Analysis of Optimization Methods and Their Applications in Real life.

University : Galgotias University Greater Noida.

PUBLICATIONS:

Published articles in reputed Journals.

- [1] Bilal, Millie Pant, **Hira Zaheer** et al. "Differential Evolution: A review of more than two decades of research, in Engineering Applications of Artificial Intelligence, Elsevier, 0952-1976, DOI 2020.103479. (2020) **SCI IF 8**
- [2] **Hira Zaheer**, Millie Pant, "Solution to Industrial Optimization Problems through Differential Evolution Variants", in Materials and Manufacturing Processes, Taylor and Francis.(pp 1-13), 2017, DOI: 10.1080/10426914.2017.1279300 **SCI IF 4.8**
- [3] Rastogi, Shubhangi, and **Hira Zaheer**. "Comparative analysis of queuing mechanisms: Droptail, RED and NLRED." Social Network Analysis and Mining 6.1 (2016): 70. DOI:10.1007/s13278-016-0382-5 **ESCI IF 2.8**
- [4] **Zaheer, H.**, Pant, M., Kumar, S., Monakhov, O., Monakhova, E., & Deep, K. (2014). A new guiding force strategy for differential evolution. International Journal of System Assurance Engineering and Management, 2170-2183, DOI 10.1007/s13198-014-0322-6. **ESCI IF 2**
- [5] **Hira Zaheer**, et. al. "Multitask Planning problem using Differential Evolution for Integer Programming" in "Cluster Computing" Springer India **SCI 4.4** [communicated]

Book Chapters

- [1] Kumar, Sushil, Peeyush Joshi, Vanita Garg, and **Hira Zaheer**. "Single Image Reflection Removal Using Deep Learning." In Design and Applications of Nature Inspired Optimization, pp. 163-177. Springer, Cham, **2022**.
- [2] **Hira Zaheer**, Millie Pant, Sushil Kumar and Oleg Monakhov (**2016**) "A Novel Mutation Strategy for Differential Evolution", Problem Solving and Uncertainty Modeling through Optimization and Soft Computing Applications, IGI Global, pp-20-31. [9781466698857]
- [3] Kumar, Sushil, **Hira Zaheer**, in "Role of Nature Inspired Algorithms in Real World Problems" Springer **2024 (Accepted)**
- [4] Vanita Garg, **Hira Zaheer**, Deepti Singh, Prashant Kumar, Om Prakash Verma "Improved Marine Predator Algorithm for real world problems" in "Role of Nature-Inspired Algorithms in Real-life Problems", Springer **2024**. (Communicated)

Published articles in International Conferences

- [1] **Hira Zaheer**, Millie Pant: "Solution of multi – objective portfolio optimization problem using Multi-Objective Synergetic Differential Evolution (MO-SDE)" in AISC series of Springer, 2017.
 - [2] Shubhangi Rastogi, **Hira Zaheer**, "Significance of parallel computation over serial computation using OpenMP, MPI and CUDA" Proceedings in Business and Economics. Springer, Singapore, (pp 359- 367), 2017.
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- [3] **Hira Zaheer**, Millie Pant, Oleg Monakhova, Emilia Monakhova, "A Simple and Efficient Co-operative Approach for Solving Multi-modal Problems", in IEEE Digital Explore Library, (pp. 4001-4005), 2016.
 - [4] **Hira Zaheer**, Millie Pant, "Solving Portfolio Optimization through Differential Evolution", in IEEE Digital Explore Library, (pp. 3892-3897), 2016.
 - [5] Shubhangi Rastogi, **Hira Zaheer**, "Significance of parallel computation over serial computation", in IEEE Digital Explore Library, (pp. 2307-2310), 2016.
 - [6] **Hira Zaheer**, Millie Pant, Oleg Monakhova, Emilia Monakhova, "A Portfolio Analysis of Ten National Banks Through Differential Evolution", in Proceedings of Fifth International Conference on Soft Computing for Problem Solving, Springer India, 2016
 - [7] **Zaheer, H.**, & Pant, M. (2015, January). "A Differential Evolution Approach for Solving Integer Programming Problems". In Proceedings of Fourth International Conference on Soft Computing for Problem Solving (pp. 409-420). Springer India.
 - [8] **Zaheer, H.**, Pant, M., Monakhov O., MonakhovaE. "Solving Portfolio Optimization Problem through Differential Evaluation" XI International Conference on Innovation and Business Management (ICIBM-15).

GOOGLE SCHOLAR DATA

https://scholar.google.co.in/citations?user=50Rg_SoAAAAJ&hl=en&authuser=1

Total Citations	689
h-index	7
i10-index	5

SCHOLARSHIP AND AWARDS:

1. UGC Junior Research fellowship during Ph.D. (2013-2014) (Rs. 25000 Per Month)
2. UGC Senior Research fellowship during Ph.D. (2015-2017) (Rs. 28000 Per Month)
3. Full air fare and stay in for a DST-DAAD Indo German project for a month in Germany.
4. Received a seed grant of Rs 1.1Lakh for the project "**Solution of a Real-Life Optimization problems using Hybrid Metaheuristics**" under Seed grant scheme of Galgotias University Greater Noida.

INTERNATIONAL CONFERENCES AND WORKSHOP ATTENDED:

1. International Conference on Electrical, Electronics and Optimization Techniques (ICEEOT), DMI College of Engineering Chennai, Tamilnadu, 3rd – 5th March 2016.
 2. 5th International Conference on "Soft Computing for Problem Solving" (SocPros 2015) at Saharanpur Campus of IIT Roorkee, 18-20 December 2015.
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3. XI International Conference on Innovation and Business Management (ICIBM) held at Hotel Grand Shiva, Shankar Ashram Chowk, Haridwar- 249407, Uttarakhand (India) during June 20-21, 2015
 4. 4th International Conference on “Soft Computing for Problem Solving” (SocPros 2014) at Department of Mathematics, NIT Silchar, Cachar, Assam, 27-29 December 2014.
 5. 3rd International Conference on “Soft Computing for Problem Solving” (SocPros 2013) at Greater Noida Extension Centre of IIT Roorkee during 26-28 December 2013.
 6. National Workshop On “Optimization Techniques & Their Applications” held at Motilal Nehru National Institute of Technology Allahabad. (June 05-11, 2013).
 7. One week DST sponsored National Training Programme on “Soft Computing Techniques for Optimization” held at ABV-Indian Institute of Information Technology & Management, Gwalior during 4-9 March, 2013.

RECOGNITION:

➤ *Reviewer for referred International conferences:*

- 12th International Conference on “Soft Computing for Problem Solving” (SocPros 2023), IIT Roorkee.
- International Conference on Computing, Communication and Automation (ICCCA2016), 29-30 April 2016.
- 5th International Conference on “Soft Computing for Problem Solving” (SocPros 2015), Saharanpur Campus, IIT Roorkee.
- International Conference on Computing, Communication and Automation (ICCCA 2015), 15-16 May 2015.
- 4th International Conference on “Soft Computing for Problem Solving” (SocPros 2014), NIT Silchar.
- 3rd International Conference on “Soft Computing for Problem Solving” (SocPros 2013), held at Greater Noida extension Centre of IIT Roorkee.

➤ *Volunteer in referred International conferences:*

- 5th International Conference on “Soft Computing for Problem Solving” (SocPros 2015), Saharanpur Campus, IIT Roorkee.
- 3rd International Conference on “Soft Computing for Problem Solving” (SocPros 2013), Greater Noida extension Centre of IIT Roorkee.

➤ *Proposed Special session in referred International conferences:*

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- 4th International Conference on “Soft Computing for Problem Solving” (SocPros 2014), NIT Silchar.
 - Proposed and chaired session in Soft Computing: Theories and Applications(SoCTA2018), NIT Jalandhar.

EXTRA-CURRICULAR ACTIVITIES:

Worked in the following at IIT Roorkee:

- Assistant warden in Indira Bhawan (Girls Hostel), for June 2015- May 2017.
- Polling Officer in election committee for year, 2015-2016, 2016-2017, 2017-2018.
- Member in hobbies club for June 2015- May 2017.
- Member in cultural club, Indira Bhawan from June 2015- May 2017.
- Invited member in Student Affair Council from June 2015- May 2017.
- Member in Anti Ragging Squads three times in academic year 2014-2015, 2015-2016, 2016-2017.

REFERENCES:

1. Prof. Millie Pant, Professor,
Department of Applied Science and Engineering, IIT Roorkee,
E-mail: millidma@gmail.com, Mobile: +919759561464
2. Dr Sushil Kumar, Assistant Professor
Department of Computer Engineering, NIT Kurukshetra,
E-mail: skumar@nitkkr.ac.in, Mobile: +919557677251

DECLARATION:

I hereby declare that the information above is true to the best of my knowledge & belief.

Place – Greater Noida (U.P)
Date – 20/06/2024



Dr Hira Zaheer
